Exercises taken from the paper [Ryc23]

Abstract

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1 Exercise 1

Show that in the approximation of keeping the diagrams with the maximal number of crosses, the correlation function $\langle \varphi(x)\omega(y)\rangle$ measures the susceptibility, i.e.

$$\partial_H \langle \phi(x)\rangle^{-}_{h+\delta(\cdot-y)} \approx \partial_H \phi^{-}_{h+\delta(\cdot-y)}$$

2 Exercise 2

Show that the condition $(y')^2 = y^2$ can be written in matrix form as $\mathbf{R}^{\mathrm{st}} g \mathbf{R} = g$ with the supertranspose $\mathbf{R}$ defined as

$$\mathbf{R} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \implies \mathbf{R}^{\mathrm{st}} = \begin{pmatrix} A^T & C^T \\ -B^T & D^T \end{pmatrix}$$

where we write $\mathbf{R}$ in the block diagonal form with $A_{d\times d}$ and $D_{2\times 2}$ blocks and $B_{d\times 2}$ and $C_{2\times d}$ blocks, and T is the ordinary transpose.

The condition $(y')^2 = y^2$ means

$$(y')^a g_{ab} (y')^b = \mathrm{R}^a_c y^c g_{ab} \mathrm{R}^b_d y^d \overset{!}{=} y^c g_{cd} y^d$$

which implies, for the metric,

$$\mathrm{R}^a_c g_{ab} \mathrm{R}^b_d = g_{cd}$$

Now we proceed with the assumption that we may write the matrix product in the standard form, that is

$$\begin{aligned} \mathbf{R}^T g \mathbf{R}^T &= g \\ \mathbf{R} &= \begin{pmatrix} A & B \\ C & D \end{pmatrix} \\ g &= \begin{pmatrix} \mathbb{1} & \mathbb{0}_{d\times 2} \\ \mathbb{0}_{2\times d} & R_\alpha \end{pmatrix} \\ R_\alpha &= \begin{pmatrix} 0 & \alpha \\ -\alpha & 0 \end{pmatrix} \end{aligned}$$

By using the above, we arrive at the following set of equations,

$$\begin{cases} A^T A + C^T R_\alpha C = \mathbb{1} \\ A^T B + C^T R_\alpha D = \mathbb{0}_{d\times 2} \\ B^T A + D^T R_\alpha C = \mathbb{0}_{2\times d} \\ B^T B + D^T R_\alpha D = R_\alpha \end{cases}$$

Since we want to be as general as possible, let's suppose that all the matrices A, B, C, D are different from $\mathbb{0}$.

We see a strange thing happening in the second and third lines. Since $R_\alpha = -R_\alpha^T$, then by transposing the third line and summing it with the second we get

$$A^T B = \mathbb{0}_{d\times 2} \implies B = \mathbb{0}_{d\times 2}$$

contradicting our hypothesis. Therefore, the above schema is wrong. The way to correct it[1] would be to render the second and third line the same (under transposition). To do this, we redefine

[1] I don't know if this procedure **unique**.

$$
\begin{aligned}
\mathbf{R}^{\mathrm{st}} g \mathbf{R}^T &= g \\
\mathbf{R}^{\mathrm{st}} &= \begin{pmatrix} A^T & C^T \\ -B^T & D^T \end{pmatrix} \\
g &= \begin{pmatrix} \mathbb{1} & \mathbb{0}_{d\times 2} \\ \mathbb{0}_{2\times d} & R_\alpha \end{pmatrix} \\
R_\alpha &= \begin{pmatrix} 0 & \alpha \\ -\alpha & 0 \end{pmatrix}
\end{aligned}
$$

Now we get this set of equations:

$$
\begin{cases}
A^T A + C^T R_\alpha C = \mathbb{1} \\
A^T B + C^T R_\alpha D = \mathbb{0}_{d\times 2} \\
-B^T A + D^T R_\alpha C = \mathbb{0}_{2\times d} \\
-B^T B + D^T R_\alpha D = R_\alpha
\end{cases}
$$

Indeed, if we transpose the third line and sum it with the second we get

$$
A^T B + C^T R_\alpha D - A^T B - C^T R_\alpha D = \mathbb{0}_{d\times 2}
$$

3 Exercise 2.5

Check that the quadratic part of the PS action gives the momentum space propagators

$$
\begin{aligned}
G_{\varphi\omega}(p) &= \frac{1}{p^2+m^2} \\
G_{\varphi\varphi}(p) &= \frac{R}{(p^2+m^2)^2} \\
G_{\omega\omega}(p) &= 0 \\
G_{\bar{\psi}\psi} &= \frac{1}{p^2}
\end{aligned}
$$

We employ the fourier transforms of both φ and ω in the massless ($m^2 = 0$) theory. Since the ghosts, at tree level, don't couple with the fields, we know that

$$G_{\bar{\psi}\psi} = \frac{1}{p^2}$$

Now, for the interesting part,

$$
\begin{aligned}
S_{\text{PS}} &= \int \left\{ -\frac{R}{2}\omega^2 + \omega\left[-\partial^2\varphi\right] \right\} d^dx \\
&= \int \left\{ -\frac{R}{2} e^{i(p+p')x}\tilde{\omega}(p)\tilde{\omega}(-p) + p^2\frac{1}{2}\left[\tilde{\omega}(p)\tilde{\varphi}(p') + \tilde{\omega}(p')\tilde{\varphi}(p)\right]\right] \right\} d^dx \frac{d^p}{(2\pi)^2}\frac{d^dp'}{(2\pi)^2} \\
&= \int \left\{ -\frac{R}{2}\tilde{\omega}(p)\tilde{\omega}(-p) + \frac{1}{2}p^2\left[\tilde{\omega}(p)\varphi(-p) + \tilde{\omega}(-p)\varphi(p)\right] \right\} \frac{d^dp}{(2\pi)^d}
\end{aligned}
$$

To see what matrix we need to invert to find the various propagators, so let's make the associations

$$
\begin{cases}
\tilde{\omega} = a_1 \\
\tilde{\varphi} = a_2
\end{cases}
$$

with this, we may rewrite the PS action as

$$S_{\text{PS}} = \int a_i G^{ij} a_j \frac{d^dp}{(2\pi)^2}$$

where the matrix G is

$$\begin{pmatrix} -\frac{R}{2} & \frac{p^2}{2} \\ \frac{p^2}{2} & 0 \end{pmatrix}$$

It's inverse is, inserting back the mass depence,

$$G^{-1} = \begin{pmatrix} 0 & \frac{2}{p^2+m^2} \\ \frac{2}{p^2+m^2} & \frac{2R}{(p^2+m^2)^2} \end{pmatrix}$$

We may as well absorb the factor of 2, since it comes from inverting the momenta $p \mapsto -p$.

We read off precisely the remaining propagators,

$$\begin{aligned}
\langle \omega\omega \rangle &= 0 \\
\langle \varphi \rangle &= \frac{1}{p^2 + m^2} \\
\langle \varphi\varphi \rangle &= \frac{R}{(p^2 + m^2)^2}
\end{aligned}$$

4 Exercise 2.8

Work out transformations of the superfield components corresponding to the super-rotations

$$\begin{cases} \delta x^\mu = \alpha(\epsilon^\mu \bar{\theta} + \bar{\epsilon}^\mu \theta) \\ \delta\theta = -\epsilon^\mu x_\mu \\ \delta\bar{\theta} = \bar{\epsilon}^\mu x_\mu \end{cases}$$

In particular, show that $\delta\omega = \alpha\left(\bar{\epsilon}^\mu \partial_\mu \psi - \epsilon^\mu \partial_\mu \bar{\psi}\right)$

Let's do the computations step by step. In the following, we keep only the terms up to $O(\epsilon^2)$. Moreover, we directly perturb both in the fields and the coordinates. For the field φ, we have

$$\varphi(x^\mu + \delta x^\mu) \approx \varphi(x^\mu) + \partial_\mu \varphi \delta x^\mu$$

For the $\bar{\theta}\psi$ field we have

$$\bar{\theta}'\psi(x^\mu + \delta x^\mu) \approx (\bar{\theta} + \delta\bar{\theta})(\psi + \partial_\mu \psi \delta x^\mu) = \bar{\theta}\psi - \alpha\theta\bar{\theta}\partial_\mu \psi \bar{\epsilon}^\mu + \bar{\epsilon}^\mu x_\mu \psi$$

Notice the very interesting fact that we get a correction not on the ghosts but on the auxilary field ω. For the $\theta\bar{\psi}$ field we have almost the same,

$$\theta\bar{\psi}(x^\mu + \delta x^\mu) \approx (\theta + \delta\theta)(\bar{\psi} + \partial_\mu \bar{\psi} \delta x^\mu) = \theta\bar{\psi} + \alpha\theta\bar{\theta}\partial_\mu \bar{\psi} \epsilon^\mu - \epsilon^\mu x_\mu \bar{\psi}$$

Lastly, for the $\theta\bar{\theta}\omega$ field we have

$$\begin{aligned} \theta'\bar{\theta}'\omega(x^\mu + \delta x^\mu) &\approx (\theta + \delta\theta)(\bar{\theta} + \delta\bar{\theta})(\omega + \partial_\mu \omega \delta x^\mu) = \theta\bar{\theta}\omega + \bar{\theta}\delta\theta\omega + \theta\delta\bar{\theta}\omega \\ &= \theta\bar{\theta}\omega + \theta\bar{\epsilon}^\mu x_\mu \omega - \bar{\theta}\epsilon^\mu x_\mu \omega \end{aligned}$$

We sum all the terms to see how the superfield Φ transform,

$$\begin{aligned} \Phi(y + \delta y) \approx \varphi &+ \epsilon^\mu \bar{x}_\mu \psi - \epsilon^\mu x_\mu \bar{\psi} \\ &+ \theta(\bar{\psi} + \alpha\partial_\mu \varphi \bar{\epsilon}^\mu + \bar{\epsilon}^\mu x_\mu \omega) \\ &+ \bar{\theta}(\psi + \alpha\partial_\mu \varphi \epsilon^\mu - \epsilon^\mu x_\mu \omega) + \\ &+ \theta\bar{\theta}(\omega + \alpha\partial_\mu \bar{\psi} \epsilon^\mu - \alpha\partial_\mu \psi \bar{\epsilon}^\mu) \end{aligned}$$

From this, we read that the infinitesimal transformations are

$$\begin{cases} \delta\varphi = \epsilon^\mu \bar{x}_\mu \psi - \epsilon^\mu x_\mu \bar{\psi} \\ \delta\bar{\psi} = \alpha\partial_\mu \varphi \bar{\epsilon}^\mu + \bar{\epsilon}^\mu x_\mu \omega \\ \delta\psi = \alpha\partial_\mu \varphi \epsilon^\mu - \epsilon^\mu x_\mu \omega \\ \delta\omega = \alpha\partial_\mu \bar{\psi} \epsilon^\mu - \alpha\partial_\mu \psi \bar{\epsilon}^\mu \end{cases}$$

5 Exercise 2.9

We note that S_{PS} can be written as a superspace integral:

$$S_{\text{PS}} = \int d^d x d\bar{\theta} d\theta \mathcal{L}$$

$$\mathcal{L} = -\frac{1}{2}\Phi(\partial_\mu \partial^\mu + R\partial_{\bar{\theta}}\partial_\theta)\Phi + V(\Phi)$$

The integral over the Grassman variables is normalized as $\int \theta\bar{\theta} d\bar{\theta} d\theta$. The task is to verify the above formula.

We do this by computing every term. Since we are dealing with Grassman variables, there are many terms which are either null by their multiplication, or by integration. To see this rewrite $\mathcal{L} - V(\Phi)$ as

$$\begin{aligned}\mathcal{L} - V(\Phi) &= -\frac{1}{2}\left[\varphi + \theta\bar{\psi} + \bar{\theta}\theta\omega\right](\partial_\mu\partial^\mu + R\partial_{\bar{\theta}}\partial_\theta)\left[\varphi + \theta\bar{\psi} + \bar{\theta}\theta\omega\right] \\ &= -\frac{1}{2}\left[\varphi + \theta\bar{\psi} + \bar{\theta}\theta\omega\right](\partial_\mu\partial^\mu(\dots) + R\omega)\end{aligned}$$

where the dots are equal to the terms in square bracket. Of these, only those proportional to $\theta\bar{\theta}$ survive, so

$$\mathcal{L} - V(\Phi) = \theta\bar{\theta}\left(\varphi\partial^2\omega + R\omega^2 + \bar{\psi}\partial_\mu\partial^\mu\psi - \psi\partial_\mu\partial^\mu\bar{\psi} + \omega\partial_\mu\partial^\mu\varphi\right)$$

Then, by doing the Grassman integration, we obtain

$$S_{\text{PS}} = -\frac{1}{2}\int\left(\varphi\partial^2\omega + R\omega^2 + \bar{\psi}\partial_\mu\partial^\mu\psi - \psi\partial_\mu\partial^\mu\bar{\psi} + \omega\partial_\mu\partial^\mu\varphi\right)d^d x + \int d^d x d\theta d\bar{\theta} V(\Phi)$$

We almost recognise the form of S_{PS} given in the (glossary?). We just need to integrate by parts the $\varphi\partial^2\omega$ term and recognise the potential term.

So we track the terms

$$\begin{aligned}\varphi\partial^2\omega &= \partial_\mu(\varphi\partial^\mu\omega - \omega\partial^\mu\varphi) + \omega\partial_\mu\partial^\mu\varphi \\ \psi\partial_\mu\partial^\mu\bar{\psi} &= \partial_\mu(\psi\partial_\mu\bar{\psi} + \bar{\psi}\partial_\mu\psi) - \bar{\psi}\partial_\mu\partial^\mu\psi\end{aligned}$$

Assuming that

$$(\varphi\partial^\mu\omega - \omega\partial^\mu\varphi)\Big|_{\partial\mathcal{M}} = 0 = (\psi\partial_\mu\bar{\psi} + \bar{\psi}\partial_\mu\psi)\Big|_{\partial\mathcal{M}}$$

We indeed see that the free theory is reproduced by the above transformation. Now we calculate the potential term. Specifically, we require that

$$\int d^d x d\theta d\bar{\theta} V(\Phi) = \int d^d x \left(\omega V'(\varphi) + \bar{\psi} V''(\varphi)\psi\right)$$

which implies

$$V(\Phi) = \theta\bar{\theta}\left(\omega V'(\varphi) + \bar{\psi} V''\psi\right)$$

6 Exercise 2.10

A radial function is a function of y^2. Check that the superLaplacian maps radial functions to radial functions.

7 Exercise 3.1

Show that the momentum-space propagator is given by

$$G_{ij} = \frac{\delta_{ij}}{p^2+m^2} + \frac{RM_{ij}}{(p^2+m^2)(p^2+m^2-nR)}$$

where the i, j indexes refer to replica i, j.

First of all, the effective action we get after integrating over $\mathcal{D}h$ is

$$S_n = \int \left[\frac{1}{2}\sum_{i=1}^{n}(\partial\phi_i)^2 + \frac{1}{2}\sum_{i=1}^{n} m^2\phi_i^2 - \frac{1}{2}R\left(\sum_{i=1}^{n}\phi_i\right)^2 \right] d^dx$$

Let's Fourier transform every replicated field, that is

$$\phi_i(x) = \int \frac{d^dp}{(2\pi)^d} e^{ipx}\tilde{\phi}_i(p)$$

The action S_n thus reads, term by term

$$\begin{aligned}
\int \sum_{i=1}^{n}(\partial\phi_i)^2 d^dx &= \sum_i \int (-pp')e^{i(p+p')x}\tilde{\phi}_i(p)\tilde{\phi}_i(p')\, d^dx \frac{d^dp}{(2\pi)^d}\frac{d^dp'}{(2\pi)^d} \\
&= \sum_i \int p^2\tilde{\phi}_i(p)\tilde{\phi}_i(-p)\, \frac{d^dp}{(2\pi)^d} \\
\int \sum_{i=1}^{n}\phi_i^2\, d^dx &= \sum_i \int \tilde{\phi}_i(p)\tilde{\phi}_i(-p)\, \frac{d^dp}{(2\pi)^d} \\
\int \left(\sum_i \phi_i\right)^2 d^dx &= \int \sum_i \phi_i^2\, d^dx + \int \sum_{i\neq j}^{n} \phi_i\phi_j\, d^dx \\
&= \sum_i \int \tilde{\phi}_i(p)\tilde{\phi}_i(-p)\, \frac{d^dp}{(2\pi)^d} + \frac{1}{2}\sum_{i\neq j}^{n}\int \left(\tilde{\phi}_i(p)\tilde{\phi}_j(-p) + \tilde{\phi}_i(-p)\tilde{\phi}_j(p)\right) \frac{d^dp}{(2\pi)^d}
\end{aligned}$$

The last term is subtle: we need to account for the interaction between replicas. Collecting every term[2], we find that

$$S_n = \sum_i^n \int \phi_i(p)\left[(p^2+m^2)\delta^{ij} - R\mathbf{B}^{ij}\right]\phi_j\, \frac{d^dp}{(2\pi)^d},$$

where $B_{ij} = 1\,\forall i, j$. From the above formula we have the inverse propagator, which is

$$G^{ij} = (p^2+m^2)\delta_{ij} - RB^{ij}$$

[2]For simplicity, I assume the various ϕ_i have the same functional form, so we can say $\frac{1}{2}\int \tilde{\phi}_i\phi_j(\tilde{-p}) + \tilde{\phi}_i(-p)\tilde{\phi}_j(p)\,\frac{d^dp}{(2\pi)^d} = \int \tilde{\phi}_i(p)\phi_j(-p)\,\frac{d^dp}{(2\pi)^d}$

To find the inverse of this matrix, make use of the this formula[3]:

$$(\mathrm{A}+\mathrm{H})^{-1} = \mathrm{A}^{-1} - \frac{1}{1+\mathrm{tr}\mathrm{H}\mathrm{A}^{-1}}\mathrm{A}^{-1}\mathrm{H}\mathrm{A}^{-1}$$

By setting $A = (p^2+m^2)\mathbb{1}$ and $H = -R\mathrm{B}$ we find

$$\begin{aligned}\Big[(p^2+m^2)\mathbb{1} - R\mathrm{B}\Big]^{-1} &= (p^2+m^2)^{-1}\mathbb{1} - \frac{-R}{1-\frac{nR}{(p^2+m^2)}}(p^2+m^2)^{-2}\mathrm{B} \\ &= (p^2+m^2)^{-1}\mathbb{1} + \frac{R}{(p^2+m^2)(p^2+m^2-nR)}\mathrm{B}\end{aligned}$$

[3]Thanks to the Sherman-Morrison formula

8 Exercise 3.4

Plug the transformation

$$\begin{cases} \phi_1 = \varphi + \frac{\omega}{2} \\ \phi_i = \varphi - \frac{\omega}{2} + \chi_i, \ \sum\limits_{i=2}^{n} \chi_i = 0 \end{cases}$$

into the quadratic part of the replicated Lagrangian $\sum\limits_{i=1}^{n} (\partial\phi_i)^2 - \frac{R}{2}\left(\sum\limits_{i=1}^{n} \phi_i\right)^2$ and see that it reproduces the quadratic part of S_{PS}, i.e. $\partial\varphi\partial\omega - \frac{R}{2}\omega^2 + \frac{1}{2}\sum(\partial\chi_i)^2$, up to terms which are $O(n)$.

Before proceeding, note that since $\sum\limits_{i=2}^{n} \chi_i = 0$, then it is true that, being this a finite sum, $\sum\limits_{i=2}^{n} \partial\chi_i = 0$. Now, being precise is key in this exercise, so we expand term by term the quadratic part of the replicated Lagrangian. For the term $\sum\limits_{i=2}^{n} \frac{1}{2}(\partial\phi_i)^2$ we get

$$\begin{aligned}
\frac{1}{2}\sum_{i=1}^{n}(\partial\phi_i)^2 &= \frac{1}{2}(\partial\phi_1)^2 + \frac{1}{2}\sum_{i=2}^{n}(\partial\phi_i)^2 = \frac{1}{2}\left[\partial\left(\varphi + \frac{\omega}{2}\right)\right]^2 + \frac{1}{2}\sum_{i=2}^{n}\left[\partial\left(\varphi - \frac{\omega}{2} + \chi_i\right)\right]^2 \\
&= \frac{1}{2}(\partial\varphi)^2 + \frac{1}{8}(\partial\omega)^2 + \frac{1}{2}\partial\varphi\partial\omega + \frac{1}{2}\sum_{i=2}^{n}(\partial\varphi)^2 + \frac{1}{4}(\partial\omega)^2 + (\partial\chi_i)^2 + \\
&+ \frac{1}{2}\sum_{i=2}^{n} -\partial\varphi\partial\omega + 2\partial\varphi\partial\chi_i - \partial\omega\partial\chi_i \\
&= \partial\varphi\partial\omega + \frac{1}{2}\sum_{i=2}^{n}(\partial\chi_i)^2 + n\left[\frac{1}{2}(\partial\varphi)^2 + \frac{1}{8}(\partial\omega^2) - \frac{1}{2}\partial\varphi\partial\omega\right] + \partial\varphi\cancelto{0}{\sum_i(\partial\chi_i)} + \\
&- \frac{1}{2}\partial\omega\cancelto{0}{\sum_i \partial\chi_i} \\
&= \partial\varphi\partial\omega + \frac{1}{2}\sum_{i=2}^{n}(\partial\chi_i)^2 + \frac{n}{2}\left[\partial\left(\varphi - \frac{\omega}{2}\right)\right]^2
\end{aligned}$$

Notice that we will get, in general, terms that are proportional to the number of replicas. These will go to 0 in the physical limit $n \to 0$.

Now, for the $\left(\sum\limits_{i=1}^{n} \phi_i\right)^2$ we have

$$
\begin{aligned}
\left(\sum_{i=1}^{n}\phi_i\right)^2 &= \sum_{i=1}^{n}\phi_i^2 + 2\sum_{i>j}^{n}\phi_i\phi_j = \phi_1^2 + \sum_{i=2}^{n}\phi_i^2 + 2\phi_1\sum_{j>1}^{n}\phi_j + 2\sum_{1\neq i>j}^{n}\phi_i\phi_j \\
&= 2\varphi\omega + n\left(\varphi - \frac{\omega}{2}\right)^2 + \sum_{i>1}(\chi_i)^2 + 2(n-1)\left(\varphi^2 - \frac{\omega^2}{4}\right) + \\
&+ n(n-1)\left(\varphi - \frac{\omega}{2}\right)^2 + 2\sum_{1\neq i>j}\chi_i\chi_j \\
&= 2\varphi\omega - 2\varphi^2 + \frac{\omega^2}{2} + 2n\left(\varphi^2 - \frac{\omega^2}{4}\right) + n^2\left(\varphi - \frac{\omega}{2}\right)^2 + \cancelto{0}{\left(\sum_i \chi_i\right)^2} \\
&= -2\left(\varphi - \frac{\omega}{2}\right)^2 + \omega^2 + 2n\left(\varphi^2 - \frac{\omega^2}{4}\right) + n^2\left(\varphi - \frac{\omega}{2}\right)^2
\end{aligned}
$$

9 Exercise 3.6

Spontaneous breaking of S_n invariance could be detected in the lack of S_n invariance of the 2-pt function $\langle \phi_i \phi_j \rangle$ for $i, j = 1, \ldots, n$. Use the Cardy transformation and the propagators

$$\begin{cases} \langle \varphi\omega \rangle = \frac{1}{p^2} \\ \langle \varphi\varphi \rangle = \frac{R}{p^4} \\ \langle \chi_i \chi_j \rangle = \frac{\delta_{ij}}{p^2} - \frac{1}{p^2}\frac{1}{n-1}\mathrm{B} \end{cases}$$

to compute the propagators $\langle \phi_1 \phi_1 \rangle$, $\langle \phi_1 \phi_i \rangle$ and $\langle \phi_i \phi_j \rangle$. Deduce that these are consistent with the S_n invariance (for $m^2 = 0$)

For this, we just need to remember that $\langle \omega\omega \rangle = 0$, as we computed in 3. The first propagator is

$$\langle \phi_1 \phi_1 \rangle = \langle \varphi\varphi \rangle + \langle \varphi\omega \rangle = \frac{R}{p^4} + \frac{1}{p^2}$$

The second propagator, $\langle \phi_1 \phi_i \rangle$, for any $i = 2, \ldots, n$ gives

$$\langle \phi_1 \phi_i \rangle = \langle \varphi\varphi \rangle + \langle \varphi\chi_i \rangle - \left\langle \frac{\omega}{2}\chi_i \right\rangle = \langle \varphi\varphi \rangle = \frac{R}{p^4}$$

Lastly, for the $\langle \phi_i \phi_j \rangle$, we compute

$$\langle \phi_i \phi_j \rangle = \langle \varphi\varphi \rangle + \langle \chi_i \chi_j \rangle = \frac{1}{p^2} + \frac{\delta_{ij}}{p^2} - \frac{1}{p^2}\frac{1}{n-1}\mathrm{B}_{ij}$$

Since both δ_{ij} and B_{ij} are invariante under index permutation, we conlcude that every propagator is S_n invariant.

10 Exercise 3.8

Reproduce the leading scaling dimensions 2Δ of the operators

$$\begin{aligned}\sigma_1\sigma_3, \sigma_2^2 &= \text{dim.}8 - 2\epsilon + \text{higher}\\ \sigma_1^2\sigma_2 &= \text{dim.}10 - 2\epsilon + \text{higher}\\ \sigma_1^4 &= \text{dim.}12 - 2\epsilon + \text{higher}\end{aligned}$$

by transforming the $\sigma_k \overset{def.}{=} \sum_i \phi_i^k$ interactions in the Cardy field basis.

The exercise consists in seeing that there are only two relevant type of blocks:

$$\begin{aligned}&\left(\varphi + \frac{\omega}{2}\right)\\ &\left(\varphi - \frac{\omega}{2} + \chi_i\right)\end{aligned}$$

if we are to check the scaling dimension of the operators above shown, we really have to check which combination of

$$\left(\varphi + \frac{\omega}{2}\right)^\alpha \left(\varphi - \frac{\omega_2}{2} + \chi_i\right)^\beta$$

has highest scaling dimension in terms of α and β.

For the interactions $\sigma_1\sigma_3$ and σ_2^2, we have the lowest dimension corresponding to $\alpha = 1$ and $\beta = 0$, that is

$$\left[\varphi + \frac{\omega}{2}\right]_d = \left(\frac{d}{2} - 1\right) \cdot 4 = 16 - 4\epsilon \implies \Delta = 8 - 2\epsilon$$

For the interaction $\sigma_1^2\sigma_2$ the relevant block is

11 Exercise 3.9

Show that the permutation

$$P_k \ : \text{index} \to \text{index} \tag{1}$$
$$\phi_1 \mapsto \phi_k \tag{2}$$
$$\tag{3}$$

acts on the Cardy fields φ, ω, χ_i as

$$\begin{aligned}
\varphi &\mapsto \varphi + (\chi_k - \omega) \\
\omega &\mapsto \omega \\
\chi_k &\mapsto 2\omega - \chi_k \\
\chi_i &\mapsto \chi_i - \chi_k + \omega
\end{aligned}$$

First of all, the permutation over n index is a group of order 2, so we know that $P_k^2 = \mathbb{1}$. With this, we consider

$$\begin{aligned}
\left(\varphi + \frac{\omega}{2}\right) &= P_k\left(\varphi + \frac{\omega}{2}\right) = P_k\left(\varphi - \frac{\omega}{2} + \chi_k\right) \\
&= P_k(\varphi) - \frac{1}{2}P_k(\omega) + P_k(\chi_k)
\end{aligned}$$

Now, the permutation will act on the fields, generally, by sending them into a linear combination of the Cardy fields. So

$$\left(\varphi + \frac{\omega}{2}\right) = \varphi(a + a' + a'') + \frac{\omega}{2}(b + b' + b'') + \chi(c + c' + c'')$$

Therefore, we have to choose the coefficients a, b, c so that the following system is satisfied:

$$\begin{cases}
a + a' + a'' = 1 \\
b + b' + b'' = 1 \\
c + c' + c'' = 0
\end{cases}$$

Choosing $\mathbf{a} = \big(1, 0, 0\big)$, $\mathbf{b} = \big(-1, 1, 2\big)$ and $\mathbf{c} = \big(1, 0, -1\big)$ we find that

$$\begin{aligned}
\varphi &\mapsto \varphi + (\chi_k - \omega) \\
\omega &\mapsto \omega \\
\chi_k &\mapsto 2\omega - \chi_k
\end{aligned}$$

For the last relation, we first notice that any element of this group is its inverse: $P_k^{-1} = P_k \, \forall k$. With this, we write

$$P_j\phi_k = P_jP_k\phi_1 = \left(P_k^{-1}P_j^{-1}\right)^{-1}\phi_1 \overset{!}{=} P_kP_j\phi_1 = P_k\phi_j$$

that is

$$(P_j - P_k)(\phi_k + \phi_j) = 0 \implies P_j(\chi_k) - P_k(\chi_j) = \chi_k - \chi_j$$

So, we can conclude that for $i \neq j$ we have $P_k(\chi_j) = -\chi_k$

12 Exercise 3.10

Show that this expression

$$\mathcal{O} = L + \sum_{k=2}^{n} P_k L$$

where L is a leader, is fully S_n invariant. With this trick, we can reconstruct any S_n invariant interaction by considering only the leaders of a theory.

To be (or not to be lol) S_n invariant means that $\mathcal{O}$ follows

$$P_k \mathcal{O} = \mathcal{O}, \forall k$$

So, we check the definitions,

$$\begin{aligned}
\mathcal{O} = L + \sum_{k=2}^{n} P_k L &\overset{!}{=} P_j L + P_j \sum_{k=2}^{n} P_k L = P_j L + \sum_{k=2}^{n} P_j P_k L \\
&= P_j L + \sum_{j \neq k}^{n} P_j P_k L + \delta_{ij} P_j P_k L = P_j^2 + P_j L + \sum_{j \neq k}^{n} P_j P_k L \\
&= L + \sum_{k=2}^{n} P_k L
\end{aligned}$$

Q.E.D

13 Exercise 3.10

If L is not a leader, then the formula $\mathcal{O} = L + \sum_{k\geq 2}^{n} P_k L$ will still generate a singlet interaction, but its leader will not be, in general, L. Show what happens in the cases where $L = \varphi^2, \varphi\omega, \omega \sum_{k=2}^{n} \chi_i^3$

Again, an exercise in computation, But first, we need to know what happens when the permutation group acts on the product of Cardy fields. (check for representation theory stuff). In general, letting $\mathbf{a} = \left(\varphi, \omega, \chi_2, \ldots, \chi_n\right)$, we have that

$$P_k \mathbf{a}^n = (P_k \mathbf{a})^n$$

In the first case, we have

$$P_k \varphi^2 = (P_k \varphi)^2 = (\varphi - \omega + \chi_k)^2$$

Then, the formula reads

$$\mathcal{O} = -\omega^2 - 2\varphi\omega + \sum_{k=2}^{n} \chi_k^2 + O(n)$$

The leader has indeed changed: now it is $\varphi\omega$. In the second case with $\varphi\omega$ we have

$$\mathcal{O} = \varphi\omega + \sum (\varphi - \omega + \chi_k)\, \omega = \omega^2 + O(n)$$

Lastly, for the case with $L = \omega \sum_{k=2}^{n} \chi_i^3$ we have

$$\mathcal{O} =$$

References

[Ryc23] Slava Rychkov. *Lectures on the random field ising model: From parisi-sourlas supersymmetry to dimensional reduction.* Springer Nature, 2023.